Sayan Ghosh, Ph.D.

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https://sg0.github.io IEEE Senior Member (# 92684851)



Employment History

2006 – 2007 **Software Engineer.** Keane, Inc, Bangalore, India.

2007 – 2009 **Software Engineer.** Thomson Reuters, Bangalore, India.

2019 – present **Computer Scientist.** Integrated Discovery Sciences Directorate, Pacific Northwest National Laboratory, Richland, WA, United States.

Education

2010 – 2014 **M.S. Computer Science, University of Houston**
Thesis title: *Energy Efficiency of Parallel Scientific Kernels*.

2015 – 2019 **Ph.D. Computer Science, Washington State University**
Thesis title: *Supporting Efficient Graph Analytics and Scientific Computation Using Asynchronous Distributed-Memory Programming Models*.

Research Publications

Journal Articles

1. M. Jain, N. B. Agostini, **S. Ghosh**, and A. Tumeo, "Analyzing inference workloads for spatiotemporal modeling," *Available at SSRN 4772671*,
2. M. Jain, N. B. Agostini, **S. Ghosh**, and A. Tumeo, "Analyzing inference workloads for spatiotemporal modeling," *Future Generation Computer Systems*, vol. 163, p. 107 513, 2025.
3. O. Subasi, **S. Ghosh**, J. Manzano, B. Palmer, and A. Marquez, "Analysis and benchmarking of feature reduction for classification under computational constraints," *Machine Learning: Science and Technology*, vol. 5, no. 2, p. 020 501, 2024.
4. N. Gawande, **S. Ghosh**, M. Halappanavar, A. Tumeo, and A. Kalyanaraman, "Towards scaling community detection on distributed-memory heterogeneous systems," *Parallel Computing*, vol. 111, p. 102 898, 2022.
5. **S. Ghosh**, N. R. Tallent, and M. Halappanavar, "Characterizing performance of graph neighborhood communication patterns," *IEEE Transactions on Parallel and Distributed Systems*, vol. 33, no. 4, pp. 915–928, 2021.
6. **S. Ghosh**, T. Liao, H. Calandra, and B. M. Chapman, "Performance of cpu/gpu compiler directives on iso/tti kernels," *Computing*, vol. 96, no. 12, pp. 1149–1162, 2014.

Conference/Workshop Proceedings

1. **S. Ghosh**, E. Song, and A. Márquez, "Adapting workloads for cxl-based disaggregated-memory systems," in *2026 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, IEEE, 2026, pp. 167–170.
2. M. Mandulak, S. Ferdous, **S. Ghosh**, M. Halappanavar, and G. M. Slota, "Approximate matching for fuzzy set similarity," in *2026 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, IEEE, 2026, pp. 813–816.

3. A. Sarkar, **S. Ghosh**, N. Tallent, A. Chadha, T. G. Roosta, and A. Jannesari, "Rudder: Steering prefetching in distributed gnn training using llm agents," in *Proceedings of the 40th ACM International Conference on Supercomputing*, 2026, pp. 353–366.

4. C. N. Avans et al., "Concepts for designing modern c++ interfaces for mpi," in *European MPI Users' Group Meeting*, Springer Nature Switzerland Cham, 2025, pp. 165–183.

5. M. Mandulak, **S. Ghosh**, S. Ferdous, M. Halappanavar, and G. Slota, "Anonymized network sensing using c++ 26 std:: Execution on gpus," in *2025 IEEE High Performance Extreme Computing Conference (HPEC)*, IEEE, 2025, pp. 1–7.

6. M. N. Newaz, **S. Ghosh**, N. R. Tallent, and G. Qu, "Locality aware process remapping for distributed-memory graph workloads," in *2025 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, IEEE, 2025, pp. 447–459.

7. Y. Kang, **S. Ghosh**, M. Kandemir, and A. Marquez, "Impact of write-allocate elimination on fujitsu a64fx," in *Proceedings of the International Conference on High Performance Computing in Asia-Pacific Region Workshops*, 2024, pp. 24–35.

8. Y. Kang, **S. Ghosh**, M. Kandemir, and A. Márquez, "Studying cpu and memory utilization of applications on fujitsu a64fx and nvidia grace superchip," in *Proceedings of the International Symposium on Memory Systems*, 2024, pp. 198–207.

9. M. Mandulak, **S. Ghosh**, S. Ferdous, M. Halappanvar, and G. Slota, "Efficient weighted graph matching on gpus," in *SC24: International Conference for High Performance Computing, Networking, Storage and Analysis*, IEEE, 2024, pp. 1–16.

10. M. N. Newaz, **S. Ghosh**, J. Suetterlein, N. R. Tallent, M. A. Mollah, and H. Ming, "Graph analytics on jellyfish topology," in *2024 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, IEEE Computer Society, 2024, pp. 839–851.

11. B. Palmer, **S. Ghosh**, and A. Márquez, "Distributed-memory sparse deep neural network inference using global arrays," in *2024 IEEE High Performance Extreme Computing Conference (HPEC)*, IEEE, 2024, pp. 1–7.

12. A. Sarkar, **S. Ghosh**, N. R. Tallent, and A. Jannesari, "Massivegnn: Efficient training via prefetching for massively connected distributed graphs," in *2024 IEEE International Conference on Cluster Computing (CLUSTER)*, IEEE, 2024, pp. 62–73.

13. H.-Y. Chou and **S. Ghosh**, "Batched graph community detection on gpus," in *Proceedings of the International Conference on Parallel Architectures and Compilation Techniques*, 2022, pp. 172–184.

14. **S. Ghosh**, "Improved distributed-memory triangle counting by exploiting the graph structure," in *2022 IEEE High Performance Extreme Computing Conference (HPEC)*, IEEE, 2022, pp. 1–6.

15. M. Jain, **S. Ghosh**, and S. P. Nandanoori, "Workload characterization of a time-series prediction system for spatio-temporal data," in *Proceedings of the 19th ACM International Conference on Computing Frontiers*, 2022, pp. 159–168.

16. H. Lee, M. Jain, and **S. Ghosh**, "Sparse deep neural network inference using different programming models," in *2022 IEEE High Performance Extreme Computing Conference (HPEC)*, IEEE, 2022, pp. 1–6.

17. **S. Ghosh**, C. Alsobrooks, M. Rüfenacht, A. Skjellum, P. V. Bangalore, and A. Lumsdaine, "Towards modern c++ language support for mpi," in *2021 Workshop on Exascale MPI (ExaMPI)*, IEEE, 2021, pp. 27–35.

18. **S. Ghosh**, Y. Guo, P. Balaji, and A. H. Gebremedhin, "Rmacxx: An efficient high-level c++ interface over mpi-3 rma," in *2021 IEEE/ACM 21st International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, IEEE, 2021, pp. 143–155.

19 **S. Ghosh**, N. R. Tallent, M. Minutoli, M. Halappanavar, R. Peri, and A. Kalyanaraman, "Single-node partitioned-memory for huge graph analytics: Cost and performance trade-offs," in *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, 2021, pp. 1–14.

20 M. Halappanavar, M. Minutoli, and **S. Ghosh**, "Graph analytics in the exascale era," in *Proceedings of the 18th ACM International Conference on Computing Frontiers*, 2021, pp. 209–209.

21 **S. Ghosh** and M. Halappanavar, "Tric: Distributed-memory triangle counting by exploiting the graph structure," in *2020 IEEE High Performance Extreme Computing Conference (HPEC)*, IEEE, 2020, pp. 1–6.

22 **S. Ghosh**, M. Halappanavar, A. Kalyanaraman, A. Khan, and A. H. Gebremedhin, "Exploring mpi communication models for graph applications using graph matching as a case study," in *2019 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, IEEE, 2019, pp. 761–770.

23 **S. Ghosh**, M. Halappanavar, A. Tumeo, and A. Kalyanarainan, "Scaling and quality of modularity optimization methods for graph clustering," in *2019 IEEE High Performance Extreme Computing Conference (HPEC)*, IEEE, 2019, pp. 1–6.

24 **S. Ghosh**, M. Halappanavar, A. Tumeo, A. Kalyanaraman, and A. H. Gebremedhin, "Minivite: A graph analytics benchmarking tool for massively parallel systems," in *2018 IEEE/ACM Performance Modeling, Benchmarking and Simulation of High Performance Computer Systems (PMBS)*, IEEE, 2018, pp. 51–56.

25 **S. Ghosh**, M. Halappanavar, A. Tumeo, A. Kalyanaraman, and A. H. Gebremedhin, "Scalable distributed memory community detection using vite," in *2018 IEEE High Performance extreme Computing Conference (HPEC)*, IEEE, 2018, pp. 1–7.

26 **S. Ghosh** et al., "Distributed louvain algorithm for graph community detection," in *32nd IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, 2018, pp. 885–895.

27 **S. Ghosh** and A. H. Gebremedhin, "Parallelization of bin packing on multicore systems," in *2016 IEEE 23rd International Conference on High Performance Computing (HiPC)*, IEEE, 2016, pp. 311–320.

28 **S. Ghosh**, J. R. Hammond, A. J. Pena, P. Balaji, A. H. Gebremedhin, and B. Chapman, "One-sided interface for matrix operations using mpi-3 rma: A case study with elemental," in *Parallel Processing (ICPP), 2016 45th International Conference on*, IEEE, 2016, pp. 185–194.

29 J. R. Hammond, **S. Ghosh**, and B. M. Chapman, "Implementing openshmem using mpi-3 one-sided communication," in *Workshop on OpenSHMEM and Related Technologies*, Springer International Publishing Cham, 2014, pp. 44–58.

30 N. Namashivayam, **S. Ghosh**, D. Khaldi, D. Eachempati, and B. Chapman, "Native mode-based optimizations of remote memory accesses in openshmem for intel xeon phi," in *Proceedings of the 8th International Conference on Partitioned Global Address Space Programming Models*, 2014, pp. 1–11.

31 P. Ghosh, J. R. Hammond, **S. Ghosh**, and B. Chapman, "Performance analysis of the nwchem tce for different communication patterns," in *International Workshop on Performance Modeling, Benchmarking and Simulation of High Performance Computer Systems*, Springer International Publishing Cham, 2013, pp. 281–294.

32 **S. Ghosh**, S. Chandrasekaran, and B. Chapman, "Statistical modeling of power/energy of scientific kernels on a multi-gpu system," in *2013 International Green Computing Conference Proceedings*, IEEE, 2013, pp. 1–6.

33 **S. Ghosh**, S. Chandrasekaran, and B. Chapman, "Energy analysis of parallel scientific kernels on multiple gpus," in *2012 Symposium on Application Accelerators in High Performance Computing*, IEEE, 2012, pp. 54–63.

34 **S. Ghosh**, T. Liao, H. Calandra, and B. M. Chapman, "Experiences with openmp, pgi, hmpp and openacc directives on iso/tti kernels," in *2012 SC Companion: High Performance Computing, Networking Storage and Analysis*, IEEE, 2012, pp. 691–700.

35 **S. Ghosh** and B. Chapman, "Programming strategies for gpus and their power consumption," in *2011 International Conference on Parallel Architectures and Compilation Techniques*, IEEE, 2011, pp. 218–218.

Skills

Languages	Strong competencies for English, Hindi, Bengali.
Coding	C, C++, CUDA and Python
Distributed-memory	Message Passing Interface (MPI), OpenSHMEM, GPU-based Collective Communication Libraries, etc.
Profilers	HPE CrayPAT, NVIDIA NSight Compute, Intel VTune, PAPI, Linux `perf`, etc.
Machine Learning	PyTorch and Amazon DistDGL
Misc.	Performance analysis/modeling, algorithms development and analysis, distributed-memory implementations, communication optimizations, integration with third-party BLAS runtimes, etc.

Service

2022–2026	IEEE Symposium on High Performance Interconnects (HotI). Vice Publications Chair for IEEE Micro Special Issue, Vice Technical Program Committee Chair, Vice General Chair and General Chair.
2025–2026	IEEE TCPP HPC Student Cohort (HPCSC) Co-Chair.
2022–2026	IEEE International Parallel & Distributed Processing Symposium (IPDPS), (2025) Student Travel Chair, Technical Program Committee: Parallel and Distributed Algorithms for Computational Science, Multidisciplinary and Algorithms tracks.
2022–2025	ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis, Technical Program: Machine Learning and HPC track, Research Posters and Birds of a Feather (BOFs) tracks.
2019–2026	Misc. journal paper reviewer and conference program committees: JPDC, TPDS, TPDS (special issue on AI/ML), JCST, PARCO, TOPC, IEEE Access, ISC, ICS, ICPP and HiPC.